

MedTrack_DV — KPI Definitions, Formulas & Calculations

Project: MedTrack_DV — Hospital Operations & Patient Analytics Dashboard **Group:** Group 2 **Mentor:** Pathina Likhita

1. Overview

The MedTrack_DV Tableau workbook contains healthcare operational KPIs covering patient volume, admissions, length of stay, occupancy, readmission, waiting time, patient satisfaction, hospital capacity and department efficiency.

The project requires a minimum of six KPIs. The Tableau workbook contains twelve KPI measures, all computed from the cleaned and enriched dataset:

`cleaned_healthcare_patient_flow_data_enriched.csv` — **9,216 patient records, 41 analytical fields, 19 months** (April 2023 – October 2024), **7 departments**.

Note on scope: KPI values in this document are computed over the full dataset. In the live Tableau dashboards, KPI values respond to the selected filters (department, date range, admission status, demographics).

2. General Notation

Symbol	Meaning
N	Number of records in the current dashboard context
COUNTD(x)	Count of distinct (unique) values of field x
COUNT(x)	Count of (non-null) records of field x
AVG(x)	Arithmetic mean of field x = $\Sigma x / N$
SUM(x)	Total of field x across records

3. Total Patients

- **Tableau worksheet:** `KPI - Total Patients`
- **Field:** `patient_id`
- **Formula:**

```
Total Patients = COUNTD(patient_id)
```

- **Definition:** Total number of unique patients represented within the selected dashboard context.
- **Purpose:** Measures overall patient volume.
- **Computed value (full dataset):** 9,216 unique patients.

4. Total Admissions

- **Tableau calculated field:** Total Admissions
- **Field:** patient_admission_flag
- **Formula:**

```
Total Admissions = COUNT(patient_admission_flag)
```

- **Definition:** Counts admission records represented by the patient admission flag.
- **Purpose:** Measures hospital admission volume.
- **Computed value (full dataset):** 9,216 admission records (4,612 Admitted + 4,604 Not Admitted).

5. Average Length of Stay (LOS)

- **Tableau worksheet:** KPI - Average LOS
- **Field:** length_of_stay_days
- **Formula:**

```
Average LOS = AVG(length_of_stay_days) =  $\Sigma$  length_of_stay_days / N
```

- **Worked calculation (full dataset):**

```
 $\Sigma$  length_of_stay_days = 25,322 days  
N = 9,216 records  
Average LOS = 25,322 / 9,216 = 2.7476  $\approx$  2.7 days
```

- **Admitted-only view (analysis view used on the dashboards):**

```
Admitted records N = 4,612  
Average LOS (admitted) = 25,322 / 4,612 = 5.4905  $\approx$  5.5 days
```

- **Purpose:** Measures patient stay duration and supports operational efficiency analysis.

6. Occupancy Rate

- **Tableau worksheet:** [KPI - Occupancy Rate](#)
- **Field:** [bed_occupancy_rate](#) (pre-enriched per-record fraction of beds occupied)
- **Formula:**

Occupancy Rate = $\text{AVG}(\text{bed_occupancy_rate}) = \Sigma \text{bed_occupancy_rate} / N$

- **Worked calculation (full dataset):**

$\Sigma \text{bed_occupancy_rate} = 7,136.725$
 $N = 9,216 \text{ records}$
 $\text{Occupancy Rate} = 7,136.725 / 9,216 = 0.7744 \rightarrow 77.4\%$

- **Purpose:** Measures hospital occupancy and capacity pressure.
- **Reference band:** observed record-level range 65% – 90%; the healthcare “capacity pressure” threshold commonly referenced is 85%.

7. Readmission Rate

- **Tableau worksheet:** [KPI - Readmission Rate](#)
- **Field:** [is_readmitted](#) (binary indicator: 1 = readmitted, 0 = not readmitted)
- **Formula:**

Readmission Rate = $\text{AVG}(\text{is_readmitted})$
 $= (\text{number of records with is_readmitted} = 1) / N$

- **Worked calculation (full dataset):**

Readmitted records = 369
 $N = 9,216$
 $\text{Readmission Rate} = 369 / 9,216 = 0.0400 \rightarrow 4.0\%$

- **Purpose:** Supports analysis of patient outcomes and hospital performance.
- **Department range:** Physiotherapy 2.2% (lowest) → Neurology 5.7% (highest).

8. Bed Utilization Rate

- **Tableau worksheet:** [KPI 5 – Bed Utilization Rate](#)
- **Field:** [bed_occupancy_rate](#)
- **Formula:**

Bed Utilization Rate = AVG(*bed_occupancy_rate*) → 77.4%

- **Equivalent cross-check from bed counts:**

Bed Utilization = SUM(*occupied_beds*) / SUM(*total_beds*)
= 1,427,345 / 1,843,200
= 0.7744 → 77.4% (matches the AVG indicator)

- **Definition:** The workbook uses the prepared bed occupancy rate as the bed utilization indicator.
- **Purpose:** Measures utilization of hospital bed capacity.

9. Total Beds

- **Tableau worksheet:** KPI 1 – Total Beds
- **Field:** *total_beds*
- **Formula:**

Total Beds = SUM(*total_beds*)

- **Definition:** Total bed-capacity measure within the selected dashboard context.
- **Computed value (full dataset):** 1,843,200 bed-records (constant capacity of 200 beds per record × 9,216 records).
- **Purpose:** Provides an indication of available hospital capacity.

10. Occupied Beds

- **Tableau worksheet:** KPI 2 – Occupied Beds
- **Field:** *occupied_beds*
- **Formula:**

Occupied Beds = SUM(*occupied_beds*)

- **Definition:** Total occupied-bed measure within the selected dashboard context.
- **Computed value (full dataset):** 1,427,345 occupied bed-records (per-record average ≈ 155 occupied beds of the 200-bed capacity).
- **Purpose:** Measures occupied hospital bed capacity.

11. Average Wait Time

- **Tableau worksheet:** KPI 3 – Average Wait Time

- **Field:** `patient_waittime`
- **Formula:**

Average Wait Time = $\text{AVG}(\text{patient_waittime}) = \Sigma \text{patient_waittime} / N$

- **Worked calculation (full dataset):**

$\Sigma \text{patient_waittime} = 324,955.0$ minutes
 $N = 9,216$ records
 Average Wait Time = $324,955.0 / 9,216 = 35.26 \approx 35.3$ minutes

- **Purpose:** Measures patient-flow efficiency and waiting-time pressure.

12. Patients Waiting > 45 Minutes

- **Tableau worksheet:** `KPI 4 – Patients Waiting >45 Min`
- **Supporting field:** `waittime_over_45min` (Yes/No indicator, derived during enrichment as `patient_waittime > 45`)
- **Formula:**

Patients Waiting >45 Min = $\text{COUNT}(\text{patient_id WHERE waittime_over_45min} = \text{"Yes"})$
 Share of patients = $\text{COUNT(Yes)} / \text{COUNTD(patient_id)}$

- **Worked calculation (full dataset):**

Records with `waittime_over_45min = "Yes"` → 2,750 patients
 Share = $2,750 / 9,216 = 0.2984 \rightarrow 29.8\%$

- **Purpose:** Highlights longer waiting times and potential patient-flow bottlenecks.
- **Related distribution (wait-time category):** High (>45 min) 59.3% · Medium (15–45 min) 28.9% · Low (<15 min) 11.7%.

13. Patient Satisfaction Score

- **Tableau worksheet:** `KPI - Satisfaction Score`
- **Field:** `patient_satisfaction_score` (1–5 scale)
- **Formula:**

Satisfaction Score = $\text{AVG}(\text{patient_satisfaction_score}) = \Sigma \text{score} / N$

- **Worked calculation (full dataset):**

Σ patient_satisfaction_score = 27,642.0
N = 9,216 records
Satisfaction Score = 27,642.0 / 9,216 = 3.00 (on a 1-5 scale)

- **Purpose:** Provides an indicator of patient experience.

14. Department Efficiency Score

- **Tableau worksheet:** Department Efficiency Score
- **Field:** department_efficiency_score
- **Formula:**

Department Efficiency Score = AVG(department_efficiency_score)

- **Worked calculation (full dataset):**

Σ department_efficiency_score = 595,979.71
N = 9,216 records
Hospital-wide average = 595,979.71 / 9,216 = 64.7

- **By department:** General Practice 71.7 · Renal 65.8 · Orthopedics 45.7 · Physiotherapy 44.2 · Gastroenterology 35.2 · Cardiology 28.4 · Neurology 0.5.
- **Purpose:** Supports comparison of operational efficiency across departments.

15. KPI Summary (with formulas)

#	KPI	Tableau Field	Formula / Aggregation	Computed Value
1	Total Patients	patient_id	COUNTD(patient_id)	9,216
2	Total Admissions	patient_admission_id	COUNT(patient_admission_id)	9,216
3	Average LOS	length_of_stay_days	AVG(length_of_stay_days)	2.7 days (5.5 admitted)
4	Occupancy Rate	bed_occupancy_rate	AVG(bed_occupancy_rate)	77.4%
5	Readmission Rate	is_readmitted	AVG(is_readmitted)	4.0%
6	Bed Utilization Rate	bed_occupancy_rate	AVG(bed_occupancy_rate)	77.4%

#	KPI	Tableau Field	Formula / Aggregation	Computed Value
7	Total Beds	total_beds	SUM(total_beds)	1,843,200 (200/record)
8	Occupied Beds	occupied_beds	SUM(occupied_beds)	1,427,345 (≈155/record)
9	Average Wait Time	patient_waittime	AVG(patient_waittime)	85.3 min
10	Patients Waiting >45 Min	waittime_over_45mi	COUNT(patient_id WHERE indicator=Yes)	2,750 (29.8%)
11	Satisfaction Score	patient_satisfaction	AVG(patient_satisfaction_score)	3.016
12	Department Efficiency Score	department_efficiency	AVG(department_efficiency)	64.7

16. KPI Usage by Dashboard

Dashboard	Main KPIs
Hospital Overview	Total Patients, Average LOS, Occupancy Rate, Readmission Rate, Satisfaction Score
Patient Flow	Total Patients, Average Wait Time, Patients Waiting >45 Min, Average LOS
Department Analytics	Total Admissions, Average Wait Time, Average LOS, Readmission Rate, Department Efficiency Score
Resource Utilization	Total Beds, Occupied Beds, Bed Utilization Rate, Average LOS, Average Wait Time

17. KPI Validation

The project evaluation target is:

KPI Accuracy > 95%

Validation performed before delivery:

- Each KPI recomputed independently from `cleaned_healthcare_patient_flow_data_enriched.csv` (9,216 records) and cross-checked against the workbook aggregates — **Pass**.

- Bed Utilization cross-check: $\text{SUM}(\text{occupied_beds}) / \text{SUM}(\text{total_beds}) = 77.4\%$ matches $\text{AVG}(\text{bed_occupancy_rate})$ — **Pass**.
 - LOS cross-check: all 25,322 stay-days belong to the 4,612 admitted records; non-admitted records carry 0 stay-days — **Pass**.
 - Dashboard interactions tested (filters, navigation, actions) — **Pass** (Module 7 QA).
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18. KPI Interpretation

KPI values depend on:

- Selected filters
- Selected time period
- Selected department
- Dataset records included in the dashboard

The KPIs are intended for healthcare operations analytics and are not clinical diagnosis or treatment measures.

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